

5.2.2.1.1. Rationale

In this use case, the user will often have the device in-hand, have immediate access to the onboarding payload, and have immediate access to the desired Commissioner.

5.2.2.1.2. High Level Flow

1. User initiates an interaction with a Commissioner.
2. User inputs the onboarding payload from the Commissionee.
3. Commissioner determines which technologies to use for [Device Discovery](#). When attempting to locate the device on IP-bearing networks, the [Commissionable Node Discovery](#) method is used and typically the DNS-SD service subtypes for long or short discriminator, and commissioning mode (see [Section 4.3.1.3, “Commissioning Subtypes”](#)) are specified to filter the results to Commissionees that match the discriminator in the onboarding payload and that are in Commissioning Mode. When attempting to locate the device via BLE advertisements, the discriminator will typically be used to filter the results. When using the NFC interface, the physical NFC tap performed by the user between the commissioner and commissionee allows device-to-device technology detection.
4. Commissioner begins the Commissioning process (see [Section 5.5, “Commissioning Flows”](#)). If more than one Commissionee is discovered, the Commissioner may further refine the results using any additional information such as a Vendor ID or Product ID that may be available in the onboarding payload. If there is still more than one discovered Commissionee, the Commissioner will typically attempt to establish a [PASE](#) secure commissioning session with each.

5.2.2.1.3. Misuse Considerations

When a device has a static onboarding payload, and the value is physically affixed to the product, it is possible for an attacker with one-time physical access to the device to obtain the onboarding payload and use it to compromise the security of the device in the future. For example, if the device is commissioned again using the same onboarding payload (for example, after a reset), then the attacker may be able to perform a person-in-the-middle attack which could result in a compromise of sensitive user data such as network credentials if passed to the device.

When a device includes device-specific information such as Vendor ID and Product ID in advertisements, then a malicious actor within advertisement range can detect this information and potentially associate it with the location of the device (and potentially, additional information about the location, such as its residents) in ways that the user did not intend.

5.2.2.2. User-Initiated Beacon Detection, Not Yet Commissioned Device

"Launch Commissioner, Discover New Devices"

In the User-Initiated Beacon Detection for a Not Yet Commissioned Device use case, the User first initiates an interaction with a Commissioner, and then indicates an intention to commission devices without providing additional information about them (no onboarding payload, etc).

5.2.2.2.1. Rationale

In this use case, the user has immediate access to a Commissioner. However, the user may not